Assignment 3: Air Quality Analysis

Your Name

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Setup

```r
library(tidyverse)
```

Data Simulation

We simulate 365 days of PM2.5 concentrations using a normal distribution, then assign each day to a season based on its day of year.

```r
set.seed(4821)

day <- seq(1, 365, 1)

# Normal distribution of PM2.5 values (negative values flipped to positive)
pm25 <- rnorm(365, mean = 20, sd = 15)
pm25 <- abs(pm25)

# Seasonal mean thresholds (not used directly in simulation but kept for reference)
seasonal_mean <- ifelse(day <= 90, 35,
                        ifelse(day <= 180, 12,
                               ifelse(day <= 270, 5,
                                      15)))

season <- ifelse(day <= 90,  "winter",
                 ifelse(day <= 180, "spring",
                        ifelse(day <= 270, "summer",
                               "fall")))
season <- factor(season, levels = c("winter", "spring", "summer", "fall"))
```

```r
Air_Quality <- data.frame(
  day = day,
  season = season,
  pm25 = pm25,
  risk = NA
)
```

Risk Classification

Each day is classified as **good** (12 μg/m^3), **moderate** (35.4 μg/m^3), or **unhealthy** (> 35.4 μg/m^3) based on EPA PM2.5 breakpoints. We also track the longest consecutive streak of unhealthy days.

```r
unhealthy_streak <- 0
max_unhealthy_streak <- 0

for (i in 1:nrow(Air_Quality)) {
  Air_Quality$risk[i] <- ifelse(Air_Quality$pm25[i] <= 12, "good",
                         ifelse(Air_Quality$pm25[i] <= 35.4, "moderate",
                                "unhealthy"))

  unhealthy_streak <- ifelse(Air_Quality$risk[i] == "unhealthy",
                             unhealthy_streak + 1,
                             0)

  max_unhealthy_streak <- ifelse(unhealthy_streak > max_unhealthy_streak,
                                 unhealthy_streak,
                                 max_unhealthy_streak)
}

Air_Quality$risk <- factor(Air_Quality$risk,
                           levels = c("good", "moderate", "unhealthy"))
```

The longest consecutive streak of unhealthy days was **4 days**.

Unhealthy Days by Season

```r
seasons <- levels(Air_Quality$season)
unhealthy_by_season <- setNames(integer(length(seasons)), seasons)

for (s in seasons) {
  season_subset <- Air_Quality[Air_Quality$season == s, ]
  unhealthy_by_season[s] <- sum(season_subset$risk == "unhealthy")
}

unhealthy_by_season
```

```
winter spring summer   fall
    15     11     12     18
```

Visualizations

PM2.5 Time Series

```r
plot1_title <- sprintf(
  "Daily PM2.5 Concentrations Over One Year (Max Unhealthy Streak: %d days)",
  max_unhealthy_streak
)

ggplot(Air_Quality, aes(x = day, y = pm25, color = season)) +
  geom_line(linewidth = 0.6, alpha = 0.8) +
  geom_hline(yintercept = 35.4, linetype = "dashed",
             color = "red", linewidth = 0.7) +
  geom_hline(yintercept = 12, linetype = "dashed",
             color = "yellow", linewidth = 0.7) +
  annotate("text", x = 355, y = 37.2, label = "Unhealthy (35.4)",
           color = "red", size = 3, hjust = 1) +
  annotate("text", x = 355, y = 13.5, label = "Moderate (12.0)",
           color = "yellow", size = 3, hjust = 1) +
  scale_color_manual(values = c(
    winter = "#5b9bd5",
    spring = "#70ad47",
    summer = "red",
    fall   = "brown"
  )) +
  labs(title = plot1_title, x = "Day of Year",
```

```
      y = "PM2.5 (µg/m³)", color = "Season") +
  theme_minimal(base_size = 12)
```

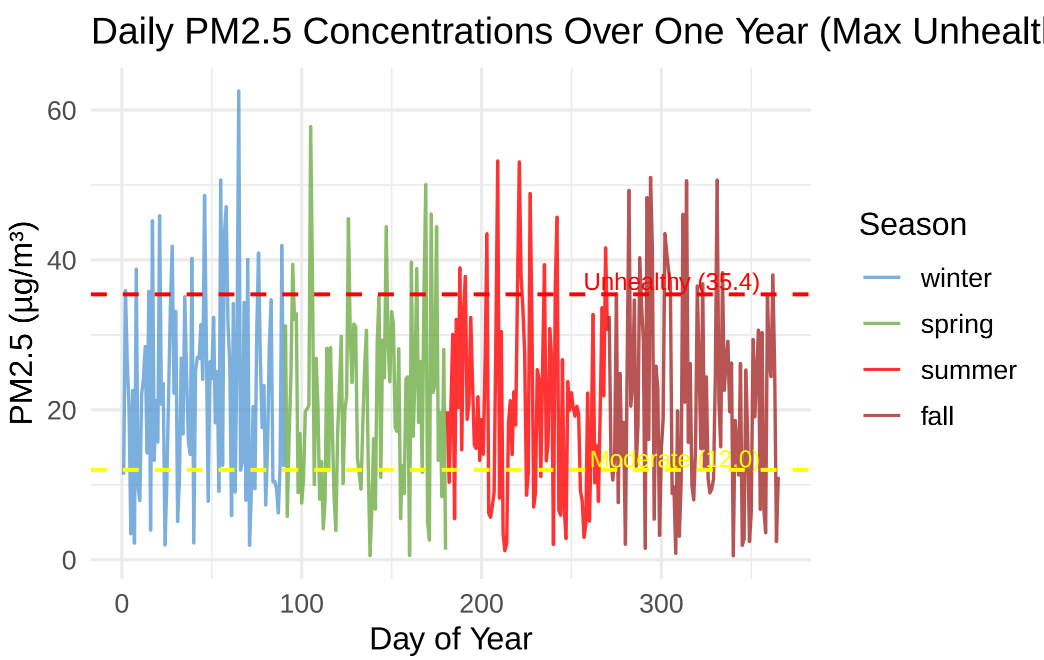

Figure 1: Daily PM2.5 concentrations over one year, colored by season. Dashed lines indicate risk thresholds.

Days per Risk Category

```
plot2_title <- sprintf(
  "Days per Air Quality Risk Category (Max Unhealthy Streak: %d days)",
  max_unhealthy_streak
)

ggplot(Air_Quality, aes(x = risk, fill = risk)) +
  geom_bar() +
  geom_text(stat = "count", aes(label = after_stat(count)),
            vjust = -0.5, size = 4.5, fontface = "bold") +
  scale_fill_manual(values = c(
    good       = "#4dac26",
    moderate   = "#f9c84a",
```

```
    unhealthy = "#d01c1f"
  )) +
  labs(title = plot2_title, x = "Risk Category", y = "Number of Days") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "none")
```

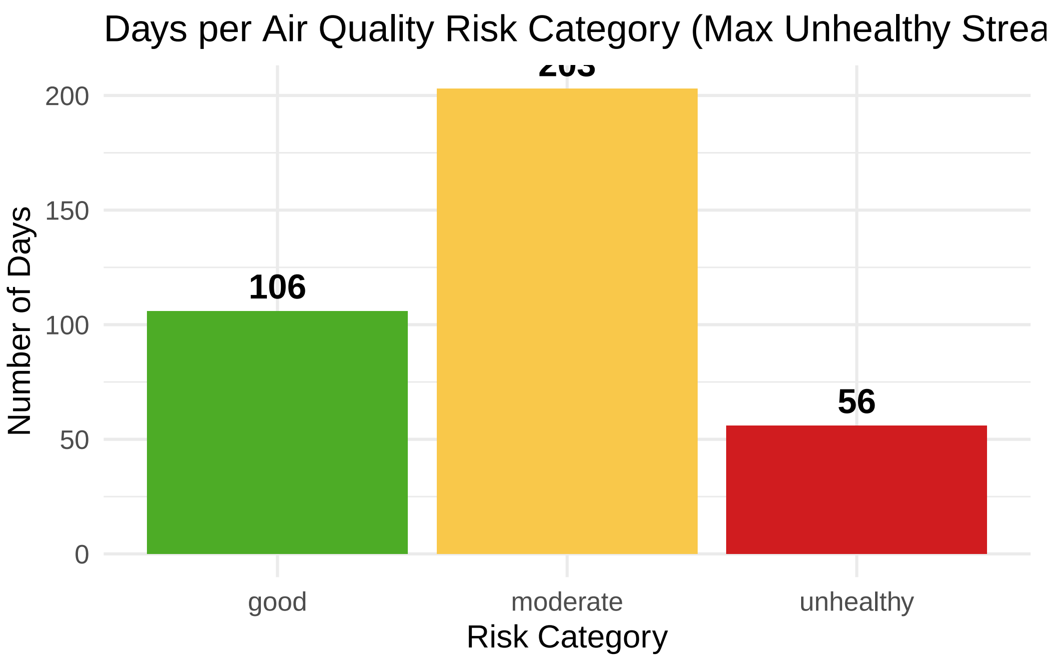

Figure 2: Count of days falling into each air quality risk category.